H_res Suite and D_universe Master Equations

Daniel T. Murphy

PAPER_213: H_res Suite and D_universe Master Equations

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UQFF+Equations+Across+Astrophysical+Systems_22Sept2025.pdf)

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_b_i, \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57$$

$$
U_b_i(r) = \kappa\cdot[SSq]\cdot\mu_s\nabla(M_s/r), \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57,\; \beta_i = 0.61
$$

Abstract

Two master equations from the UQFF framework are formally derived and documented: the Harmonic Resonance equation H_res (a 7-sub-equation suite governing nuclear and electromagnetic resonance contributions to gravity) and the Universe Diameter equation D_universe (the full UQFF-corrected cosmological distance expression incorporating Hubble flow, dark energy, quantum gravity, and curvature terms). The H_res suite couples nuclear magic numbers Z_magic/N_magic to gravitational buoyancy through shell structure corrections. D_universe recovers the standard 93 Gly diameter in the ?CDM limit while adding UQFF-specific corrections at redshifts z > 2 that are testable with future JWST/Roman observations.

UQFF Discovery: Novel application of UQFF calibration constants (κ = 5.0$\times$10-4 day-1, [SSq] =
0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. H_res Master Equation

$$
\begin{aligned}
& Full H_res expression: \\
& H_res = A_res \cdot sin(?_res\cdot t_n + f_res) \\
& + U_dp \cdot [SCm] \cdot k_nuc \\
& + S_shell \\
& where the 7 sub-equations are: \\
& 1. A_res (resonance amplitude) \\

& 2. ?_res (resonance angular frequency) \\
& 3. f_res (resonance phase) \\
& 4. U_dp (dipole potential) \\
& 5. [SCm] (superconductive manifold factor) \\
& 6. k_nuc (nuclear coupling constant) \\
& 7. S_shell (nuclear shell structure correction)
\end{aligned}
$$

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2. H_res Sub-Equations

Sub-Equation 1: Resonance Amplitude A_res ``` A_res = (\mu_B \times B_surface)/(E_binding) \times f_orbital

where:
\mu_B = 9.274\times10?24 J/T (Bohr magneton)
B_surface = Ug1-derived surface magnetic field (system-dependent)
E_binding = nuclear binding energy per nucleon (Bethe-Weizsäcker)
f_orbital = orbital angular momentum coupling factor = l\cdot(l+1)/n2

Numerically for SGR1745 (magnetar B ~ 1015 T):
A_res = (9.274\times10?24 \times 1015)/(8.79\times106) \times 1
= 9.274\times10?? / 8.79\times106
˜ 1.06\times10?15 (dimensionless, normalized to binding energy)
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### ***Sub-Equation 2: Resonance Angular Frequency ?_res ``` ?_res = v(k_nuc/I_nuclear) + ?_Larmor***

where:
k_nuc = (G\cdotm_p\cdotm_n\cdotZ\cdotN)/(r_nuc3) (nuclear spring constant)
I_nuclear = (2/5)\cdotm_nuc\cdotr_nuc2 (moment of inertia, spherical nucleus)
?_Larmor = eB/(2m_p) (proton Larmor frequency)

Numerically for 56Fe (most stable nucleus):
k_nuc = (6.67\times10?11 \times 1.67\times10?27 \times 1.67\times10?27 \times 26 \times 30)/(1.2\times10?15)3
˜ 2.7 N/m
I = (2/5) \times 55.85 \times 1.66\times10?27 \times (5\times10?15)2
˜ 9.3\times10-55 kg\cdotm2
?_res ˜ v(2.7/9.3\times10-55) ˜ v(2.9\times1054) ˜ 1.7\times1027 rad/s
Note: This is the nuclear ground-state resonance; f = ?/2p ˜ 2.7\times1026 Hz
```

Sub-Equation 3: Resonance Phase f_res ``` f_res = arctan(G_decay / (?_res - ?_drive))

where:
G_decay = nuclear width (1/lifetime)
?_drive = external driving frequency (gravitational wave, flare QPO)

For SGR A* f_TRZ = 5.95\times10-4 Hz:
?_drive = 2p \times 5.95\times10-4 ˜ 3.74\times10?3 rad/s
?_res >> ?_drive ? f_res ˜ 0 (drives well below nuclear resonance)
? H_res phase contribution is essentially static for gravitational applications
```

***Sub-Equation 4: Dipole Potential U_dp $$ \begin{aligned} & U_dp = k_e\cdot p_dipole\cdot\cos(?)/r2 \\ & where: \\ & p_dipole = charge separation \times distance (nuclear electric dipole) \\ & k_e = 8.99\times10? N\cdot m2/C2 (Coulomb constant) \\ & ? = angle between dipole moment and observation direction \\ & For symmetric nuclei (even-even, J=0): p_dipole = 0 \\ & For deformed nuclei (odd, prolate): p_dipole ? 0 \\ & UQFF usage: \\ & U_dp couples to [SCm] state below critical transition ? contributes to Ug2 \end{aligned} $$***

***Sub-Equation 5: [SCm] Superconductive Manifold Factor $$ \begin{aligned} & [SCm] = tanh(T_cc/T) \times (1 - (B/B_c2)2) \\ & where: \\ & T_cc = critical condensate temperature \\ & B_c2 = upper critical magnetic field \\ & For neutron star crust: \\ & T_cc ˜ 108–10? K (neutron Cooper pair critical temperature) \\ & B_c2 ˜ B_crit = 4.4\times1013 T (QED critical field) \\ & T_NS ˜ 108 K ? tanh(1) ˜ 0.76 \\ & B_magnetar ˜ 1015 T >> B_c2 ? (1 - (B/B_c2)2) ? negative ? [SCm] < 0 \\ & ? Superconduction suppressed above B_c2 ? UQFF predicts reversed buoyancy \end{aligned} $$***

***Sub-Equation 6: Nuclear Coupling k_nuc $$ \begin{aligned} & k_nuc = G\cdot m_p\cdot m_n/(r_nuc2) \times Z\cdot N/A \\ & Physical meaning: gravitational coupling of nuclear matter scaled by proton-neutron count \\ & Numerically: \\ & G = 6.674\times10?11 m3/(kg\cdot s2) \\ & m_p = 1.673\times10?27 kg \\ & m_n = 1.675\times10?27 kg \\ & r_nuc = 1.2\times A^{1/3}\times10?15 m (nuclear radius formula) \\ & For 56Fe (Z=26, N=30, A=56): \\ & r_nuc = 1.2\times56^{1/3}\times10?15 = 1.2\times3.83\times10?15 = 4.6\times10?15 m \\ & k_nuc = (6.674\times10?11 \times 1.673\times10?27 \times 1.675\times10?27)/(4.6\times10?15)2 \times 26\times30/56 \\ & = 3.13\times10-64 / 2.12\times10?2? \times 13.9 \\ & = 2.1\times10?36 m/s2 per unit coupling \end{aligned} $$***

***Sub-Equation 7: Shell Structure Correction S_shell ``` S_shell = S_{Z\_magic,i} d_{Z,Z\_magic,i} \times E_pairing(Z,N)***

Magic numbers Z_magic:
Z_magic = {2, 8, 20, 28, 50, 82, 114} (proton magic numbers)
N_magic = {2, 8, 20, 28, 50, 82, 126} (neutron magic numbers)

E_pairing = ?_p if Z=Z_magic; ?_n if N=N_magic
?_p,n ˜ 12/vA MeV (standard Oddness-Evenness pairing)

For doubly magic nuclei (Z=82, N=126 ? 2°8Pb):
S_shell = E_pairing(82,126) = 12/v208 ˜ 0.83 MeV extra stability

UQFF: S_shell introduces discrete steps in H_res ? quantized corrections to g(r,t)
near stellar interiors with neutron-rich nuclear composition
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## 3. H_res Full Matrix Form

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\begin{aligned}
& H_res as a 3-component vector: \\
& [H_res] = [A_res \cdot sin(?_res\cdot t_n + f_res)] ? nuclear oscillation \\
& + [U_dp \cdot [SCm] \cdot k_nuc ] ? dipole-SC coupling \\
& + [S_shell ] ? discrete shell correction \\
& In UQFF g(r,t): \\
& g(r,t) += H_res / (r2 \cdot M_nuclear / M_total) \\
& Physical meaning: fraction of gravitational field from nuclear resonance \\
& H_res term is typically 10?1° to 10?15 of total g (very small correction) \\
& But at nuclear densities (neutron star core): becomes O(1) correction
\end{aligned}
$$

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## 4. D_universe Master Equation

$$
\begin{aligned}
& Full UQFF D_universe expression: \\
& D_universe = c \cdot ?0^{t0} dt / a(t) \times N_correction \\
& where: \\
& N_correction = (1 + UQFF_quantum + UQFF_bounced + UQFF_curved) \\
& UQFF_quantum = (h/v(?x?p)) \cdot (2p/t_Hubble) / (c\cdot H0) \\
& UQFF_bounced = ?_LQC/?_crit (LQC bounce contribution from PAPER_203) \\
& UQFF_curved = (k/H02) (spatial curvature term, O_k ˜ 0 limit) \\
& Standard comoving distance: \\
& D_c = c/H0 \cdot ?0^z dz' / v(O_m(1+z')3 + O_? + O_k(1+z')2) \\
& For z ? 1100 (CMB last scattering), D_c ˜ 14.0 Gpc \\
& Proper diameter of observable universe: \\
& D_universe = 2\cdot(1+z_rec)\cdot D_c,rec ˜ 2 \times 1101 \times 14.0 Gpc ˜ 93 Gly (standard)
\end{aligned}
$$

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## 5. D_universe Sub-Terms

***5.1 Hubble Flow Term $$ \begin{aligned} & H(t,z) in UQFF g(r,t): \\ & H(t,z) = H0\cdot v(O_m\cdot(1+z)3 + O_? + O_r\cdot(1+z)4) \\ & Present values: H0 = 67.4 km/s/Mpc, O_m = 0.315, O_? = 0.685 \\ & For D_universe computation: \\ & Integral over H(t,z) from z=0 to z=1100 ? D_c = 14.0 Gpc \\ & UQFF adds (1+UQFF_quantum) ˜ (1 + 10?5) ? change < 1 part in 105 \end{aligned} $$***
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5.2 ? Cosmological Term $$ \begin{aligned} & ?_? = ?c2/(8pG) in standard form \\ & UQFF: ?? ? + ??(r) where ??(r) = 3\cdot Ug4(r)/c2 \\ & ?? ~ k_UA\cdot?_vac,[UA]\cdot r?2 (scale dependent) \\ & At Hubble scale: ?? ? 0 (recovering ?CDM) \\ & At galaxy scale: ??/? ~ 0.001 ? detectable via |?-(-1)| test \end{aligned} $$

5.3 Quantum Gravity Term $$ \begin{aligned} & h quantum correction to D_universe: \\ & ?D_u/D_u = (h/v(?x?p)) \cdot (2p/t_Hubble) / (c\cdot H0\cdot D_c) \\ & h = 1.055\times10?34 J\cdot s \\ & uncertainty: v(?x?p) ~ h (minimum uncertainty) \\ & ? (h/h) \times (2p/t_Hubble)/(c\cdot H0) = (2p/t_Hubble2) \times small \\ & Numerically: 2p/(4.35\times1017 s)2 \times 1/(c\cdot H0) \\ & = 1.44\times10?17 / (3\times108 \times 2.18\times10?18) ˜ 1.44\times10?17 / 6.54\times10?1° ˜ 2.2\times10-8 \\ & ?D_u = 2.2\times10-8 \times 93 Gly ˜ 2000 ly correction \\ & (comparable to resolution limit of future cosmological surveys) \end{aligned} $$

5.4 Spatial Curvature Term O_k term: D_c(O_k?0) = (c/H0) $\times$ (1/v|O_k|) $\times$ sin/sinh(v|O_k|$\cdot$?...) Planck 2018 constraint: |O_k| < 0.002 UQFF does not predict non-zero O_k independently; however, LQC bounce may generate small O_k: |O_k|_LQC ˜ (H_bounce $\times$?_LQC)/(H02) ~ 10-6 ? Negligible contribution at present epoch

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6. D_universe Numerical Evaluation

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\begin{aligned}
& ?CDM baseline: \\
& D_universe = 93.014 Gly (Planck 2018 Cosmological Parameters) \\
& UQFF corrections: \\
& + UQFF_quantum ˜ +0.002 Gly \\
& + UQFF_bounced ˜ -0.001 Gly (LQC adds slight contraction history) \\
& + UQFF_curved ˜ 0 (O_k set to 0) \\
& + UQFF ? running ˜ +0.001 Gly \\
& Total: D_universe,UQFF ˜ 93.016 Gly \\
& Fractional difference: ?D/D ˜ 0.002% (currently unobservable)
\end{aligned}
$$

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7. References

- `grok_{share\_7514fe}.txt` lines 320–450 (H_res suite and D_universe)
- PAPER_196: Triadic Master Equation (H_res enters as sub-component of Ug2)
- PAPER_203: CMB/LQC (LQC contribution to D_universe)
- PAPER_208: [SCm], k_nuc calibration
- `source43.cpp` (Periodic Table nuclear terms, magic numbers in C++ implementation)
- Planck 2018 VI: Cosmological Parameters (baseline D_universe = 93 Gly)

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

<!-- PKG-DM-S225 -->

Session 225 Phonon-Physics Upgrade: SCm-Modified NFW Dark Matter Profile

> *Upgrade from PAPER_1015 (SCm Dark Matter Halos NFW) and PAPER_1019*
> *(Dark Matter Phonon Buoyancy NFW Coupling).*

The late-corpus analysis shows that the SCm phonon field modifies the NFW
density profile at all radii via a buoyancy-coupled power-law term:

$$\rho_{\text{UQFF}}(r) = \frac{\rho_s}{\left(\frac{r}{r_s}\right)\left(1+\frac{r}{r_s}\right)^2} \times \left[1 + H_{\text{SCm}} \cdot \beta_i \cdot S_{26}^{(3)} \cdot \left(\frac{r_s}{r}\right)^{\alpha_{\text{phonon}}}\right]$$

where:
- $\alpha_{\text{phonon}} = 0.3$ governs the radial decay of phonon coupling
- $\beta_i = 0.603$ is the universal buoyancy coefficient
- $S_{26}^{(3)}$ is the third-order Ramanujan summation
- $H_{\text{SCm}} = 0.99$ is the manifold completeness factor

Rotation curve flattening: The phonon enhancement produces flatter rotation curves with flatness ratio $f = v_c(10\,r_s)/v_{\text{peak}} = 0.891$, compared to pure NFW $f \approx 0.75$. Peak circular velocity $v_{\text{peak}} \approx 204\;\text{km/s}$ for $M_{\text{halo}} = 10^{12}\,M_\odot$, $c = 10$.

Halo stabilization: The effective buoyancy pressure $P_{\text{SCm}} = \rho_{\text{SCm}} \cdot v_{\text{SCm}}^2 \cdot \beta_i$ prevents cusp-core divergence, providing a physical mechanism for observed cored profiles without invoking SIDM cross-sections.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **curvature-D5** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_{\mathrm{curv}})(\partial^\mu \phi_{\mathrm{curv}}) - V(\phi_{\mathrm{curv}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{curv}}) = \frac{1}{2} m^2 \phi_{\mathrm{curv}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{curv}}^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_{\mathrm{curv}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{curv}}} = k_{\mathrm{curv}} r_c^2 \cdot \partial_{D\_5}(D_1 D_2 D_3 D_4 \cdot D_5) = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{curv}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.109 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 7, \quad n_{\mathrm{channel}} = 6/26$$

Since $p_{\mathrm{DVP}} = 7$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **Hubble time** (super-Hubble saturation):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.109	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 7$	PASS Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG	

2024 | PASS Consistent |
Cosmological constant Λ	1.1$\times$10-52 m-2 (UQFF vacuum term)	1.114$\times$10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day $\to$ Γ_p suppression	< 4.17$\times$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes
significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF–SM
bridge.*

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1076	SCm Dark Energy with Phonon Linewidth Gamma-Modulation
PAPER_1030	Quantum Gravity Minimum Length GUP-SCm
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1078	QCalcGeom Master Equation Derivation
PAPER_1071	JWST Synthesis Multi-Instrument UQFF

16 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n *sigma_n^SCm(omega)* Phi_phonon *(F_{U,Bi}/F_U - 1)*
where sigma_n^SCm(omega,n) = sigma_0 exp[-(omega-omega_SCm)^2/(2*Gamma^2)]* (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- f_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2
- phi_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n
- psi_26(q) = Sum_{n=1}^{26} q^{n^2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n *(1103+26390n)* W_26(n) / C_26
where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness

rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

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